Introduction of programming :

It is a process of giving instructions to the computer to perform specific task is called as programming.

Three levels of programming.

1. Binary level programming : Binary - level programming uses only 0s & 1s. It is the language directly understood by the computer.
   Ex : 10101010 , 00001111
2. Mid / Assembly level (Assembly Language) : Assembly language uses short codes or instructions to communicate with computer hardware. It is easier to understand than binary language.
   ex : MOV. A , B. - moves the value from B to A.
3. High - level language : Use English - like words & are easier for human to read, write & understand.
   Example : python. : print ("Hello")

Introduction of python :

Python is a high - level language, interpreted programming language that is easy to read, write and learn.
It was created by Guido van Rossum. & first released in 1991.

Features of python :

* easy to learn & simple : python has a simple syntax that is easy to understand and write.
* High - level language : python uses English - like words. So programmers can write code without dealing directly with machine - level instructions.
* Interpreted Language : Python executes code through an interpret - er, making it easier to test and debug programs.
* free & open source : python programs can run on different platform independent. Operating systems. Such as windows, linux, & macOS, with little or no modifications.
* free & open source : python is freely available, & its source code can be used, modified, & distributed under its license.

* Large Standard Library : Python provides many built-in modules & functions for tasks such as mathematics, file handling & data operations.

Applications of python :

- web application
- gaming application
- desktop application
- Data Analytics
- Data Science
- AI
- Automation
- IoT

Etc...

History of python :

- python was created by Guido Van Rossum.
- python development began in December 1989. python was first released in 1991.
- python was inspired by a Programming language called ABC.
- The name python was inspired by the British comedy show Monty python's Flying Circus, not the snake.
- python 1.0 : Released in 1994, with features such as functions & exception handling.
- python 2.0 : Released in 2000, introducing features such as list comprehensions & garbage collection improvements.
- python 3.0 : Released in 2008, with important improvements to the language. It was not fully backward-compatible with python 2.
- python became popular because it is simple to learn, readable & useful for many applications.
- Today, python is widely used in Data Analytics, Artificial Intelligence, Machine Learning, web Development, & Automation.

Difference methods of print :

1. Comma (,) Method :- The comma method is used to print multiple values in a single print() statement. python automatically adds a space b/w the values.

Ex : name = "sanjana"
age = 22
print ("My name is", name, "and my age is", age)

o/p
my name is sanjana and my age is 22.

2. format() Method : the format() method is used to insert variable values into placeholders {} inside a string.

Ex : name = "sanjana"
age = 22
print("My name is {} and my age is {}". format(name, age))

o/p
My name is sanjana and my age is 22.

3. f-string Method : the f-string method is used to insert variable values directly into a string by placing variables inside curly braces {}. the string must start with the letter f.

Ex : name = "sanjana"
age = 22
print(f"My name is {name} and my age is {age}")

o/p
My name is sanjana and my age is 22.